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Numerical approaches for solving complex order monkeypox mathematical model

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ABSTRACT

The monkeypox virus (MPXV) is what causes monkeypox (MPX) disease, which is comparable to both smallpox and cowpox. Using classical, fractional-order and complex order differential equations, we offer a deterministic mathematical model of the monkeypox virus in this study to research its possible breakouts in United States. The complex order derivative makes the fractional order derivative and the integer order derivative more common when the imaginary part of the complex order equals zero and when the real part in complex order derivatives is zero in this case, the new behaviour appears that doesn't appear in integer and fractional order derivatives. Eight nonlinear complex order differential equations make up this model consisting of humans and rodents population sizes. The population of humans N_h is divided into five different classes. The population of rodent N_r is divided into three classes, and the derivatives are described in the Atangana-Baleanu-Caputo sense and Mittag-Leffler kernels are employed. Numerical methods to simulate complex order systems such as The standard and nonstandard two-step Lagrange interpolation methods are utilised to fit the model. The basic reproduction number of the model is given. The stability of the suggested model's disease-free equilibrium point is shown. Finally, we study the duration and monkeypox outbreak's transmission pattern in the United States from June 13 to Sep 16, 2022, numerical simulations to illustrate our findings are presented. The results show that keeping diseased people apart from the general population decreases the spread of disease.

1. Introduction

The monkeypox virus is the rare viral zoonic disease known as monkeypox [13]. It was initially identified in 1958 when two cases of a disease similar to the pox were discovered in primates kept for studies. The variola virus is related to the virus that causes monkeypox, which is responsible for smallpox. It was originally identified in 1970 in central Africa ([5], [6]). Fever, rash, and lymphadenopathy are some of its signs. In severe situations, carriers may get secondary bacterial infections, pneumonitis, encephalitis, and keratitis that threatens their vision. Those who are immunocompromised, pregnant, or who are children have the chance of developing monkeypox more severely. Monkeypox cases had been reported prior to the 2022 outbreak in a number of central and western African countries. The majority of cases of monkeypox outside of Africa were linked to either imported animals or international travel to nations where the illness typically occurs. These incidents were noted on various mainlands [1]. Scientists and academics have put forth a number of theoretical and mathematical studies to examine the dynamics of monkeypox [2]. In their article [4], the ecological niche and geographic distribution of human monkeypox in Africa were investigated by the authors. Researchers looked at [7] to see how many outbreaks of monkeypox there will be and how long they will last in 2022. In nations where the disease is not endemic. The patterns of monkeypox transmission during the 2022 epidemics were investigated by the authors in ([8], [9]). The authors of [10] used mathematical modelling to simulate the dynamics of the monkeypox viral infection, including treatment and immunisation methods. The monkeypox pandemic in 2022 has a stochastic model proposed in [11] with the cross-infection hypothesis in a highly disturbed environment. In [12], a novel model of monkeypox transmission that incorporates human-to-human transmission was created. The authors of [14] used fractional derivatives to create a mathematical model of monkeypox transmission.

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Fractional calculus is increasingly being used in numerous fields of research and engineering ([20],[21],[23],[26]). Fractional calculus has recently seen a number of innovative applications in the research of numerous real-world issues ([18], [25], [27], [32]). More specifically, there are a number of recent papers that deal with the modelling of deadly epidemics. The history or memory of the variable is uniquely captured by fractional derivatives ([28], [29]). It is difficult to use integer order derivatives. Usage of models of fractional derivative is justified for resolving a variety of problems since they are more accurate than integer order derivative models in terms of real data. Atangana-Baleanu Caputo (ABC) recently defined a modified Caputo fractional derivative by replacing the non-singular and nonlocal kernel with a generalised Mittag-Leffler function ([27], [43–49]). These current versions have been utilised in a variety of disciplines to represent real-world applications.

In comparison to fractional and complex order derivatives, integer-order derivatives are unable to take into consideration systems that are impacted by inherited qualities of materials and processes and historical memory ([19],[41]).

It is acknowledged that the complex order derivative makes the fractional order derivative and the integer order derivative more common when the imaginary part of the complex order equals zero ([24], [42]) and when the real part in complex order derivatives is zero in this case new behaviour appears that doesn't appear in integer and fractional order derivatives.

New definitions for the complex order derivative have recently been put out by writers in an effort to address some problems with the singularity of the kernel operators and the locality of the kernel [27]. Atangana and Baleanu [40] provide a new complex order derivative in 2020 that is based on the generalised Mittag-Leffler function as a non-local and nonsingular kernel.

The fundamental theorem of fractional calculus and the two-step Lagrange polynomials are recently merged in ([30],[39]), Based on this concept, we provide the numerical techniques for Mittag-Leffler kernel simulation of complex order differential operators.

The fundamental accomplishment of this work is the creation of a useful numerical method for the complex order monkeypox model's solutions can be roughly approximated in [15]. The nonstandard two-step Lagrange interpolation method (NS2LIM) improves the standard two-step Lagrange interpolation method (S2LIM). Numerical simulations for the complex order model are available.

The sections of the article are as follows: Complex order definitions are presented in Section 2. Section 3 provides a description of the monkeypox model in a complex order differential equations. In section 4, presented the stability of the suggested model's disease-free equilibrium point. In section 5 NS2LIM is presented. In section 6, gives a discussion of the numerical strategy utilising NS2LIM. The applicability and efficacy of the NS2LIM are shown by numerical simulations in section 7. In section 8 conclusions are found.

2. Basic notations

Definition 2.1. For complex order derivatives $z \in \mathbb{C}$, Caputo's complex order derivative for a function $K(t)$ is given as follows [31]:

$$
{}_0^C D_t^z K(t) = \begin{cases} {}_0^C D_t^{z-\lceil Re(z)\rceil}\, {}_0^C D_t^{\lceil Re(z)\rceil} K(t)\,, \quad Re(z) \in \mathbb{R}^+ \;\; and \;\; z \notin \mathbb{N}, \\ \frac{d^z K(t)}{dt^z}, \;\; if \;\; z \in \mathbb{N}, \\ {}_0^C D_t^{z-1} K(t), \;\; if \;\; Re(z) = 0, Im(z) \neq 0, \\ \frac{1}{\Gamma(1-z)} \int_0^t \frac{K'(t)}{(t-r)} dr, \;\; if \;\; Re(z), Im(z) \in \mathbb{R}, \\ K(t), \;\; if \;\; z = 0, \\ \int_0^t \frac{(t-r)^{-z-1}}{\Gamma(-z)} K(r) dr, \;\; if \;\; Re(z) \in \mathbb{R}^-. \end{cases} \tag{1}
$$

The following is the gamma function for $z \in \mathbb{C}$ [22]:

$$
\Gamma(z) = (2\pi)^{\frac{1}{2}} e^{-z} z^{z-\frac{1}{2}} \left[O\left(\frac{1}{z}\right) + 1 \right], \;\; |z| \longrightarrow \infty, \; |arg(z)| < \pi. \tag{2}
$$

Definition 2.2. For complex order derivatives $z \in \mathbb{C}$, the Grünwald-Letinkov formula (GL) for a function $K(t)$ is given as follows:

$$
{}_b^{GL} D_t^z K(t) = \lim_{h \longrightarrow 0} \frac{1}{h^z} \sum_{j=0}^{[\frac{t-b}{h}]} (-1)^j \binom{z}{j} K(t - jh), \tag{3}
$$

the integer portion of $\frac{t-b}{h}$ is denoted by $[\frac{t-b}{h}]$, the bounds of operation for ${}_b D_t^z K(t)$ are b and t, see [31] for more information.

Definition 2.3. Atangana-Baleanu complex order derivative (ABC) in the Liouville-Caputo senses is defined as follows: [20]:

$$
{}_0^{ABC} D_t^z K(t) = \frac{B(z)}{2\pi i(1-z)} \int_0^t E_z\Big[-z \frac{(t-r)^z}{1-z} \Big] K'(r) dr, \;\; z \in \mathbb{C}, \tag{4}
$$

where $Re(z) > 0$ and $B(z) = \frac{z}{\Gamma(z)} - z + 1$, and E_z is the Mittag-Leffler function:

$$
E_z(\xi) = \sum_{\psi=0}^{\infty} \frac{\xi^\psi}{\Gamma(z\psi + 1)}, \quad \xi, \; z \in \mathbb{C}.
$$

The following is the general form of the complex order ordinary differential equation:

$$
{}_0^{ABC} D_t^z K(t) = \delta(t, K), \quad K(0) = K_0, \quad z \in \mathbb{C}. \tag{5}
$$

Table 1
Model's variables [15].

Variable	Definition
S_h	Susceptible humans class
E_h	Exposed humans class
I_h	Infected humans class
Q_h	Isolation humans class
R_h	recovery humans class
S_r	Susceptible rodent class
E_r	Exposed rodent class
I_r	Infected rodent class

Table 2
Model's parameters [15].

Parameter	Value	Definition
A_h	0.029	the recruitment rate of humans
μ_h	1.5	human mortality rates at natural causes
A_r	0.2	the recruitment rate of rodents
μ_r	0.002	the number of rodents that die naturally
β_1	0.00025	the rate of human-rodent interaction
β_2	0.00006	the contact rate between humans
β_3	0.027	the contact rate between rodents
p	0.83	the recovery rate
γ_h	0.2	the disease induced death rates of humans
γ_r	0.5	the disease induced death rates of rodents
ϕ	2	the proportion of people who are still undiagnosed after being treated
α_1	0.2	the rate of human transfer from infectious class exposure
α_2	2	the rate of suspected case identification
τ	0.52	the transitional rate from isolated to recovered class

3. The mathematical model

The complex order monkeypox model is discussed in this section. This model suggested by Peter et al. in ([14], [15]). This model consists of eight nonlinear complex order differential equations consisting of humans and rodents population sizes. The population of humans N_h is divided into five different classes namely, susceptible class $S_h(t)$, exposed class $E_h(t)$, infected class $I_h(t)$, isolation class $Q_h(t)$, and recovery class $R_h(t)$. The population of rodent N_r is divided into three classes namely susceptible class $S_r(t)$, exposed class $E_r(t)$ and infected class $I_r(t)$. Table 1 lists the definitions of each variable in the proposed model. The parameters and their meaning for the proposed model are also introduced in Table 2. The complex order monkeypox model is given as follows:

$$
\begin{aligned}
&\frac{1}{2}({}_0^{ABC}D_t^{s+it} + {}_0^{ABC}D_t^{s-it})S_h(t) = A_h - \frac{(\beta_1 I_r + \beta_2 I_h)S_h}{N_h} - \mu_h S_h + \phi Q_h,\\
&\frac{1}{2}({}_0^{ABC}D_t^{s+it} + {}_0^{ABC}D_t^{s-it})E_h(t) = \frac{(\beta_1 I_r + \beta_2 I_h)S_h}{N_h} - (\alpha_1 + \alpha_2 + \mu_h)E_h,\\
&\frac{1}{2}({}_0^{ABC}D_t^{s+it} + {}_0^{ABC}D_t^{s-it})I_h(t) = \alpha_1 E_h - (\mu_h + \gamma_h + p)I_h,\\
&\frac{1}{2}({}_0^{ABC}D_t^{s+it} + {}_0^{ABC}D_t^{s-it})Q_h(t) = \alpha_2 E_h - (\phi + \tau + \mu_h + \gamma_h)Q_h,\\
&\frac{1}{2}({}_0^{ABC}D_t^{s+it} + {}_0^{ABC}D_t^{s-it})R_h(t) = pI_h + \tau Q_h - \mu_h R_h,\\
&\frac{1}{2}({}_0^{ABC}D_t^{s+it} + {}_0^{ABC}D_t^{s-it})S_r(t) = A_r - \frac{\beta_3 S_r I_r}{N_r} - \mu_r S_r,\\
&\frac{1}{2}({}_0^{ABC}D_t^{s+it} + {}_0^{ABC}D_t^{s-it})E_r(t) = \frac{\beta_3 S_r I_r}{N_r} - (\mu_r + \alpha_3)E_r,\\
&\frac{1}{2}({}_0^{ABC}D_t^{s+it} + {}_0^{ABC}D_t^{s-it})I_r(t) = \alpha_3 E_r - (\mu_r + \gamma_r)I_r.
\end{aligned} \tag{6}
$$

Following are the initial conditions [3]: $S_h(0) = 10000,\ E_h(0) = 50,\ I_h(0) = 0.57,\ Q_h(0) = 0.43,\ R_h(0) = 0,\ S_r(0) = 1000,\ E_r(0) = 100,\ I_r(0) = 10$.

The equilibrium point of a dynamical model is proved in details in [14]. The basic reproduction number of the model (6) is given as follows:

$$
\begin{aligned}
R_0 &= max(R_0^h, R_0^r),\\
R_0 &= max\Big(\frac{(\alpha_1 + \alpha_2)\alpha_1 A_h}{N_h(\alpha_1 + \alpha_2 + \mu_h)(S_h + \alpha_2 + \mu_h + P)\mu_h}, \frac{\alpha_3 A_r \tau}{N_r \mu_r^2(\mu_r + \alpha_3)}\Big),
\end{aligned}
$$

where R_0^h is the basic reproduction number of human and R_0^r is the basic reproduction of rodent. If $R_0 > 1$, this mean that the infection will be able to spread in a population. But if $R_0 < 1$, the infection will disappear.

4. Stability of the disease-free equilibrium point

The point at which there is no illnesses in the population is known as the Monkey Pox-Free equilibrium. All classes that are infected will have a value of 0 [14]. The following shows a locally asymptotically stable Monkey Pox-Free equilibrium point. Then the Monkey Pox-Free equilibrium point is [14]

$$\delta_{MFE} = (\frac{A_h}{\mu_h}, 0, 0, 0, 0, \frac{A_r}{\mu_r}, 0, 0).$$

The Jacobian matrix evaluated at δ_{MFE} in [14] then the eigenvalues are given as follows:

$$\begin{aligned}
\psi_1 &= -\frac{\mu_r^2 N_r + N_r \alpha_3 \mu_r + \sqrt{\mu_r^2 \alpha_3^2 N_r^2 + 4\alpha_3 N_r \mu_r \tau A_r}}{2\mu_r N_r}, \\
\psi_2 &= -\frac{\mu_r^2 N_r + N_r \alpha_3 \mu_r - \sqrt{\mu_r^2 \alpha_3^2 N_r^2 + 4\alpha_3 N_r \mu_r \tau A_r}}{2\mu_r N_r}, \\
\psi_3 &= -\frac{2N_r \mu_h^2 + N_r \mu_h \tau + (N_r(\alpha_1 + \alpha_2) + N_r S_h + N_r + p)\mu_h + \sqrt{W}}{2\mu_h N_r}, \\
\psi_4 &= -\frac{2N_r \mu_h^2 + N_r \mu_h \tau + (N_r(\alpha_1 + \alpha_2) + N_r S_h + N_r + p)\mu_h - \sqrt{W}}{2\mu_h N_r}, \\
\psi_5 &= -S_r - \mu_h - \tau, \\
\psi_6 &= -\mu_r, \\
\psi_7 &= -\mu_h, \\
\psi_8 &= -\mu_h.
\end{aligned} \tag{7}$$

Where

$$\begin{aligned}
W = {} & N_r^2 \mu_h^2 P^2 + 4N_r(\alpha_1 + \alpha_2)\mu_h \alpha_1 A_h - 2(N_r^2(\alpha_1 + \alpha_2) - N_r^2(\alpha_1 + \alpha_2) - N_r^2 S_h - N_r^2 \alpha_2)\mu_h^2 P \\
& + (N_r^2(\alpha_1 + \alpha_2)^2 - 2N_r^2 S_h^2 + N_r^2 \alpha_2^2 - 2(N_r^2(\alpha_1 + \alpha_2) - N_r^2 S_h)\alpha_2)\mu_h^2.
\end{aligned}$$

Therefore, the Monkey Pox-free equilibrium (MFE) is asymptotically stable if [14]

$$\frac{\alpha_3 A_r \tau}{N_r^2(\alpha_3 + \mu_r)\mu_r^2} < 1,$$

and

$$\frac{\alpha_1(\alpha_1 + \alpha_2)A_h}{N_r(\alpha_1 + \alpha_2 + \mu_h)(S_h + \alpha_2 + \mu_h + P)\mu_h} < 1.$$

5. Non-standard two-step Lagrange interpolation method

In numerical analysis, Lagrange polynomials are applied for polynomial interpolation. For a specified collection of points (x_i, y_i) with no two x_i values equal, the Lagrange polynomial is the polynomial of lowest degree that assumes at each value x_i the corresponding value y_i, so that the functions match at each point.

The initial value problem is [17]:

$$\begin{aligned}
& {}_0^{ABC}D_t^z \delta(t) = K(t, \delta(t)), \quad 0 < t \le T, \quad Re(z) > 0, \quad z \in \mathbb{C}, \\
& \delta(0) = \delta_0.
\end{aligned} \tag{8}$$

Because of fractional calculus's fundamental theorem to (4), we have

$$\delta(t) - \delta(0) = \frac{1}{2\pi i}(\frac{1-z}{B(z)} K(t, \delta(t)) + \frac{z}{B(z) + \Gamma(z)} \int_0^t K(\theta, \delta(\theta))(t-\theta)^{z-1} d\theta). \tag{9}$$

At t_{s+1}, we have

$$\begin{aligned}
\delta(t_{\psi+1}) - \delta(0) = {} & \frac{1}{2\pi i}(\frac{\Gamma(z)(1-z)}{z + \Gamma(z)(1-z)} K(t_\psi, \delta(t_\psi)) + \frac{z}{\Gamma(z) + z(1 - \Gamma(z))} \\
& \times \sum_{i=0}^{\psi} \int_{t_\psi}^{t_{i+1}} K(\theta, x(\theta))(t_{\psi+1} - \theta)^{z-1} d\theta).
\end{aligned} \tag{10}$$

Using

$$P_k(\theta) \simeq \frac{K(t_i, \delta_i)}{h}(\theta - t_{i-1}) - \frac{K(t_{i-1}, \delta_{i-1})}{h}(\theta - t_i). \tag{11}$$

When we insert Eq. (11) into (10) and follow the same steps as in [16] we obtain:

$$\begin{aligned}\delta_{\psi+1}(t) &= \delta(0) + \frac{1}{2\pi i}\left(\frac{(1-z)\Gamma(z)}{z+\Gamma(z)(1-z)}K(t_\psi,\delta(t_\psi)) + \frac{1}{z+(1-z)(z+1)\Gamma(z)}\right.\\ &\times\sum_{i=0}^{\psi}\left(h^z K(t_i,\delta_i)\big((\psi+1-i)^z(\psi-i+z+2)-(\psi-i)^z(\psi-h\right.\\ &\left.+2+2z)\right) - h^z K(t_{i-1},\delta_{i-1})\Big((\psi+1-i)^{z+1}-(\psi-i)^z\\ &\left.\left.\times(\psi-i+z+1)\Big)\right)\right).\end{aligned} \tag{12}$$

We employed a simple modulation in (12) to get good stability [38]. Changing the step size h is the purpose of this modulation to $\phi(h)$ as a result $\phi(h) = h + O(h^2)$, $0 < \phi(h) \le 1$. For more details see ([33], [34], [35], [36], [37]).

$$\begin{aligned}\delta_{\psi+1}(t) &= \delta(0) + \frac{1}{2\pi i}\left(\frac{\Gamma(z)(1-z)}{z+\Gamma(z)(1-z)}K(t_\psi,\delta(t_\psi)) + \frac{1}{z+(1-z)(z+1)\Gamma(z)}\right.\\ &\times\sum_{i=0}^{\psi}\left(\phi(h)^z K(t_i,\delta_i)\big((\psi+1-i)^z(\psi-i+z+2)-(\psi-i)^z(\psi-i\right.\\ &\left.+2+2z)\right) - \phi(h)^z K(t_{i-1},\delta_{i-1})\Big((\psi+1-i)^{z+1}-(\psi-h)^z\\ &\left.\left.\times(\psi-i+z+1)\Big)\right)\right).\end{aligned} \tag{13}$$

6. Numerical discretization

In order to numerically discretize the complex order derivative, we create the NS2LIM with ABC complex order derivative in this part. The system's (6) numerical representation using the NS2LIM (13) is as follows:

$$\begin{aligned}
&{}^{ABC}_{0}D_t^{s+it}S_h(t) = K_1, \quad {}^{ABC}_{0}D_t^{s+it}E_h(t) = K_2, \quad {}^{ABC}_{0}D_t^{s+it}I_h(t) = K_3,\\
&{}^{ABC}_{0}D_t^{s+it}Q_h(t) = K_4, \quad {}^{ABC}_{0}D_t^{s+it}R_h(t) = K_5, \quad {}^{ABC}_{0}D_t^{s+it}S_r(t) = K_6,\\
&{}^{ABC}_{0}D_t^{s+it}S_r(t) = K_7, \quad {}^{ABC}_{0}D_t^{s+it}I_r(t) = K_8,\\
&{}^{ABC}_{0}D_t^{s-it}S_h(t) = K_1^*, \quad {}^{ABC}_{0}D_t^{s-it}E_h(t) = K_2^*, \quad {}^{ABC}_{0}D_t^{s-it}I_h(t) = K_3^*,\\
&{}^{ABC}_{0}D_t^{s-it}Q_h(t) = K_4^*, \quad {}^{ABC}_{0}D_t^{s-it}R_h(t) = K_5^*,\\
&{}^{ABC}_{0}D_t^{s-it}S_r(t) = K_7^*, \quad {}^{ABC}_{0}D_t^{s-it}I_r(t) = K_8^*,\\
&1 < s \le 2, t \in \mathbb{R}^+,
\end{aligned}$$

where

$$\begin{aligned}
&K_j = K(t, S_h, E_h, I_h, Q_h, R_h, S_r, E_r, I_r),\\
&K_j^g := K(t, S_h^g, E_h^g, I_h^g, Q_h^g, R_h^g, S_r^g, E_r^g, I_r^g),\\
&K_j^m := K(t, S_h^m, E_h^m, I_h^m, Q_h^m, R_h^m, S_r^m, E_r^m, I_r^m),\\
&K_g^* = K^*(t, S_h, E_h, I_h, Q_h, R_h, S_r, E_r, I_r),\\
&K_j^{*^m} := K^*(t, S_h^m, E_h^m, I_h^m, Q_h^m, R_h^m, S_r^m, E_r^m, I_r^m),\\
&K_j^{*^g} := K^*(t, S_h^g, E_h^g, I_h^g, Q_h^g, R_h^g, S_r^g, E_r^g, I_r^g),\\
&j = 1, \ldots, 8.
\end{aligned}$$

$$\begin{aligned}
S_{h_{\psi+1}}(t) &= S_h(0) + \frac{1}{4\pi i} * \left(\frac{\Gamma(s+it)(1-(s+it))}{\Gamma(s+it)(1-(s+it))+(s+it)}K_1^\psi + \right.\\
&\frac{2}{(s+it)+(1-(s+it))((s+it)+1)\Gamma(s+it)}\sum_{g=0}^{\psi}\Big(K_1^g\phi(h)^{s+it}\big((1-g+\psi)^{s+it}\\
&\times(\psi-g+2+(s+it)) - (\psi-g+2+2(s+it))(\psi-g)^{s+it}\big) - \phi(h)^{s+it}\\
&\left.\times K_1^{g-1}\big((\psi+1-g)^{(s+it)+1} - (\psi-g)^{(s+it)}(\psi-g+1+(s+it))\big)\Big)\right) +\\
&\left(\frac{\Gamma(s-it)(1-(s-it))}{\Gamma(s-it)(1-(s-it))+(s-it)}K_1^{*^\psi} + \frac{1}{\Gamma(s-it)(1-(s-it))((s-it)+1)+(s-it)}\right.\\
&\times\sum_{g=0}^{\psi}\Big(\phi(h)^{s-it}K_1^{*^g}\phi(h)^{s-it}\big((1-g+\psi)^{s-it}(\psi+2-g+(s-it)) - (\psi-g)^{s-it}(\psi-g+2
\end{aligned}$$

$$+2(s-it))\big)-\phi(h)^{s-it}K_1^{*^{g-1}}\left((\psi+1-g)^{(s-it)+1}-(\psi-g)^{(s-it)}(\psi-g+1+(s-it))\right)\Bigg)\Bigg)*(1/4\pi i). \tag{14}$$

$$\begin{aligned}
E_{h_{\psi+1}}(t)=E_h(0)+\frac{1}{4\pi i}*\Bigg(&\frac{\Gamma(s+it)(1-(s+it))}{\Gamma(s+it)(1-(s+it))+(s+it)}K_2^{\psi}+\\
&\frac{2}{(s+it)+(1-(s+it))((s+it)+1)\Gamma(s+it)}\sum_{g=0}^{\psi}\Bigg(K_2^{g}\phi(h)^{s+it}\big((1-g+\psi)^{s+it}\\
&\times(\psi-g+2+(s+it))-(\psi-g+2+2(s+it))(\psi-g)^{s+it}\big)-\phi(h)^{s+it}\\
&\times K_2^{g-1}\left((\psi+1-g)^{(s+it)+1}-(\psi-g)^{(s+it)}(\psi-g+1+(s+it))\right)\Bigg)\Bigg)+\\
&\Bigg(\frac{\Gamma(s-it)(1-(s-it))}{\Gamma(s-it)(1-(s-it))+(s-it)}K_2^{*^{\psi}}+\frac{1}{\Gamma(s-it)(1-(s-it))((s-it)+1)+(s-it)}\\
&\times\sum_{g=0}^{\psi}\Bigg(\phi(h)^{s-it}K_2^{*^{g}}\phi(h)^{s-it}\big((1-g+\psi)^{s-it}(\psi+2-g+(s-it))-(\psi-g)^{s-it}(\psi-g+2\\
&+2(s-it))\big)-\phi(h)^{s-it}K_2^{*^{g-1}}\left((\psi+1-g)^{(s-it)+1}-(\psi-g)^{(s-it)}(\psi-g+1+(s-it))\right)\Bigg)\Bigg)*(1/4\pi i).
\end{aligned} \tag{15}$$

$$\begin{aligned}
I_{h_{\psi+1}}(t)=I_h(0)+\frac{1}{4\pi i}*\Bigg(&\frac{\Gamma(s+it)(1-(s+it))}{\Gamma(s+it)(1-(s+it))+(s+it)}K_3^{\psi}+\\
&\frac{2}{(s+it)+(1-(s+it))((s+it)+1)\Gamma(s+it)}\sum_{g=0}^{\psi}\Bigg(K_3^{g}\phi(h)^{s+it}\big((1-g+\psi)^{s+it}\\
&\times(\psi-g+2+(s+it))-(\psi-g+2+2(s+it))(\psi-g)^{s+it}\big)-\phi(h)^{s+it}\\
&\times K_3^{g-1}\left((\psi+1-g)^{(s+it)+1}-(\psi-g)^{(s+it)}(\psi-g+1+(s+it))\right)\Bigg)\Bigg)+\\
&\Bigg(\frac{\Gamma(s-it)(1-(s-it))}{\Gamma(s-it)(1-(s-it))+(s-it)}K_3^{*^{\psi}}+\frac{1}{\Gamma(s-it)(1-(s-it))((s-it)+1)+(s-it)}\\
&\times\sum_{g=0}^{\psi}\Bigg(\phi(h)^{s-it}K_3^{*^{g}}\phi(h)^{s-it}\big((1-g+\psi)^{s-it}(\psi+2-g+(s-it))-(\psi-g)^{s-it}(\psi-g+2\\
&+2(s-it))\big)-\phi(h)^{s-it}K_3^{*^{g-1}}\left((\psi+1-g)^{(s-it)+1}-(\psi-g)^{(s-it)}(\psi-g+1+(s-it))\right)\Bigg)\Bigg)*(1/4\pi i).
\end{aligned} \tag{16}$$

$$\begin{aligned}
Q_{h_{\psi+1}}(t)=Q_h(0)+\frac{1}{4\pi i}*\Bigg(&\frac{\Gamma(s+it)(1-(s+it))}{\Gamma(s+it)(1-(s+it))+(s+it)}K_4^{\psi}+\\
&\frac{2}{(s+it)+(1-(s+it))((s+it)+1)\Gamma(s+it)}\sum_{g=0}^{\psi}\Bigg(K_4^{g}\phi(h)^{s+it}\big((1-g+\psi)^{s+it}\\
&\times(\psi-g+2+(s+it))-(\psi-g+2+2(s+it))(\psi-g)^{s+it}\big)-\phi(h)^{s+it}\\
&\times K_4^{g-1}\left((\psi+1-g)^{(s+it)+1}-(\psi-g)^{(s+it)}(\psi-g+1+(s+it))\right)\Bigg)\Bigg)+\\
&\Bigg(\frac{\Gamma(s-it)(1-(s-it))}{\Gamma(s-it)(1-(s-it))+(s-it)}K_4^{*^{\psi}}+\frac{1}{\Gamma(s-it)(1-(s-it))((s-it)+1)+(s-it)}\\
&\times\sum_{g=0}^{\psi}\Bigg(\phi(h)^{s-it}K_4^{*^{g}}\phi(h)^{s-it}\big((1-g+\psi)^{s-it}(\psi+2-g+(s-it))-(\psi-g)^{s-it}(\psi-g+2\\
&+2(s-it))\big)-\phi(h)^{s-it}K_4^{*^{g-1}}\left((\psi+1-g)^{(s-it)+1}-(\psi-g)^{(s-it)}(\psi-g+1+(s-it))\right)\Bigg)\Bigg)*(1/4\pi i).
\end{aligned} \tag{17}$$

$$\begin{aligned}
R_{h_{\psi+1}}(t)=R_h(0)+\frac{1}{4\pi i}*\Bigg(&\frac{\Gamma(s+it)(1-(s+it))}{\Gamma(s+it)(1-(s+it))+(s+it)}K_5^{\psi}+\\
&\frac{2}{(s+it)+(1-(s+it))((s+it)+1)\Gamma(s+it)}\sum_{g=0}^{\psi}\Bigg(K_5^{g}\phi(h)^{s+it}\big((1-g+\psi)^{s+it}\\
&\times(\psi-g+2+(s+it))-(\psi-g+2+2(s+it))(\psi-g)^{s+it}\big)-\phi(h)^{s+it}\\
&\times K_5^{g-1}\left((\psi+1-g)^{(s+it)+1}-(\psi-g)^{(s+it)}(\psi-g+1+(s+it))\right)\Bigg)\Bigg)+\\
&\Bigg(\frac{\Gamma(s-it)(1-(s-it))}{\Gamma(s-it)(1-(s-it))+(s-it)}K_5^{*^{\psi}}+\frac{1}{\Gamma(s-it)(1-(s-it))((s-it)+1)+(s-it)}\\
&\times\sum_{g=0}^{\psi}\Bigg(\phi(h)^{s-it}K_5^{*^{g}}\phi(h)^{s-it}\big((1-g+\psi)^{s-it}(\psi+2-g+(s-it))-(\psi-g)^{s-it}(\psi-g+2
\end{aligned}$$

$$+2(s-it))\big)-\phi(h)^{s-it}K_5^{*^{g-1}}\big((\psi+1-g)^{(s-it)+1}-(\psi-g)^{(s-it)}(\psi-g+1+(s-it))\big)\Big)\Big)*(1/4\pi i). \tag{18}$$

$$\begin{aligned}S_{r_{\psi+1}}(t)=S_r(0)+\frac{1}{4\pi i}*\Bigg(&\frac{\Gamma(s+it)(1-(s+it))}{\Gamma(s+it)(1-(s+it))+(s+it)}K_6^{\psi}+\\&\frac{2}{(s+it)+(1-(s+it))((s+it)+1)\Gamma(s+it)}\sum_{g=0}^{\psi}\Bigg(K_6^{g}\phi(h)^{s+it}\big((1-g+\psi)^{s+it}\\&\times(\psi-g+2+(s+it))-(\psi-g+2+2(s+it))(\psi-g)^{s+it}\big)-\phi(h)^{s+it}\\&\times K_6^{g-1}\big((\psi+1-g)^{(s+it)+1}-(\psi-g)^{(s+it)}(\psi-g+1+(s+it))\big)\Bigg)\Bigg)+\\&\Bigg(\frac{\Gamma(s-it)(1-(s-it))}{\Gamma(s-it)(1-(s-it))+(s-it)}K_6^{*^{\psi}}+\frac{1}{\Gamma(s-it)(1-(s-it))((s-it)+1)+(s-it)}\\&\times\sum_{g=0}^{\psi}\Bigg(\phi(h)^{s-it}K_6^{*^{g}}\phi(h)^{s-it}\big((1-g+\psi)^{s-it}(\psi+2-g+(s-it))-(\psi-g)^{s-it}(\psi-g+2\\&+2(s-it))\big)-\phi(h)^{s-it}K_6^{*^{g-1}}\big((\psi+1-g)^{(s-it)+1}-(\psi-g)^{(s-it)}(\psi-g+1+(s-it))\big)\Bigg)\Bigg)*(1/4\pi i).\end{aligned} \tag{19}$$

$$\begin{aligned}E_{r_{\psi+1}}(t)=E_r(0)+\frac{1}{4\pi i}*\Bigg(&\frac{\Gamma(s+it)(1-(s+it))}{\Gamma(s+it)(1-(s+it))+(s+it)}K_7^{\psi}+\\&\frac{2}{(s+it)+(1-(s+it))((s+it)+1)\Gamma(s+it)}\sum_{g=0}^{\psi}\Bigg(K_7^{g}\phi(h)^{s+it}\big((1-g+\psi)^{s+it}\\&\times(\psi-g+2+(s+it))-(\psi-g+2+2(s+it))(\psi-g)^{s+it}\big)-\phi(h)^{s+it}\\&\times K_7^{g-1}\big((\psi+1-g)^{(s+it)+1}-(\psi-g)^{(s+it)}(\psi-g+1+(s+it))\big)\Bigg)\Bigg)+\\&\Bigg(\frac{\Gamma(s-it)(1-(s-it))}{\Gamma(s-it)(1-(s-it))+(s-it)}K_7^{*^{\psi}}+\frac{1}{\Gamma(s-it)(1-(s-it))((s-it)+1)+(s-it)}\\&\times\sum_{g=0}^{\psi}\Bigg(\phi(h)^{s-it}K_7^{*^{g}}\phi(h)^{s-it}\big((1-g+\psi)^{s-it}(\psi+2-g+(s-it))-(\psi-g)^{s-it}(\psi-g+2\\&+2(s-it))\big)-\phi(h)^{s-it}K_7^{*^{g-1}}\big((\psi+1-g)^{(s-it)+1}-(\psi-g)^{(s-it)}(\psi-g+1+(s-it))\big)\Bigg)\Bigg)*(1/4\pi i).\end{aligned} \tag{20}$$

$$\begin{aligned}I_{r_{\psi+1}}(t)=I_r(0)+\frac{1}{4\pi i}*\Bigg(&\frac{\Gamma(s+it)(1-(s+it))}{\Gamma(s+it)(1-(s+it))+(s+it)}K_8^{\psi}+\\&\frac{2}{(s+it)+(1-(s+it))((s+it)+1)\Gamma(s+it)}\sum_{g=0}^{\psi}\Bigg(K_8^{g}\phi(h)^{s+it}\big((1-g+\psi)^{s+it}\\&\times(\psi-g+2+(s+it))-(\psi-g+2+2(s+it))(\psi-g)^{s+it}\big)-\phi(h)^{s+it}\\&\times K_8^{g-1}\big((\psi+1-g)^{(s+it)+1}-(\psi-g)^{(s+it)}(\psi-g+1+(s+it))\big)\Bigg)\Bigg)+\\&\Bigg(\frac{\Gamma(s-it)(1-(s-it))}{\Gamma(s-it)(1-(s-it))+(s-it)}K_8^{*^{\psi}}+\frac{1}{\Gamma(s-it)(1-(s-it))((s-it)+1)+(s-it)}\\&\times\sum_{g=0}^{\psi}\Bigg(\phi(h)^{s-it}K_8^{*^{g}}\phi(h)^{s-it}\big((1-g+\psi)^{s-it}(\psi+2-g+(s-it))-(\psi-g)^{s-it}(\psi-g+2\\&+2(s-it))\big)-\phi(h)^{s-it}K_8^{*^{g-1}}\big((\psi+1-g)^{(s-it)+1}-(\psi-g)^{(s-it)}(\psi-g+1+(s-it))\big)\Bigg)\Bigg)*(1/4\pi i).\end{aligned} \tag{21}$$

7. Numerical results

In the following, numerical simulations of monkeypox model in complex order derivatives are presented. Table 3 describes how the numerical algorithms NS2LIM, S2LIM, and Ode45 behave in terms of convergence at $0.9 \pm 0i$, we observed NS2LIM is convergent. The CPU time shows in Table 4 when the real part is 0.9 and the imaginary part is 0.2. To clarify the importance of the proposed model, we compared the obtained results with the real data for United States from June 13 to Sep 16, 2022. The methods NS2LIM and S2LIM are used to study the proposed model. Comparative studies using data from United States versus the simulation of the proposed model (6) at different complex order values are given in

Table 3
Comparisons between NS2LIM, S2LIM, Ode45 and $t_{final} = 1000$ at different various of h, $s = 1$, $t = 0$.

h	NS2LIM	S2LIM	Ode45
0.2	Convergent	Convergent	Convergent
15	Convergent	Divergent	Divergent
25	Convergent	Divergent	Divergent
40	Convergent	Divergent	Divergent

Table 4
The CPU time at $s = 0.98$, $t = 0.1$.

t_{final}	NS2LIM CPU time	S2LIM CPU time
800	0.4768 sec	1.3715 sec
1500	0.32554 sec	3.662143 sec
4000	0.67263 sec	5.623446 sec
5000	0.637221 sec	23.432265 sec

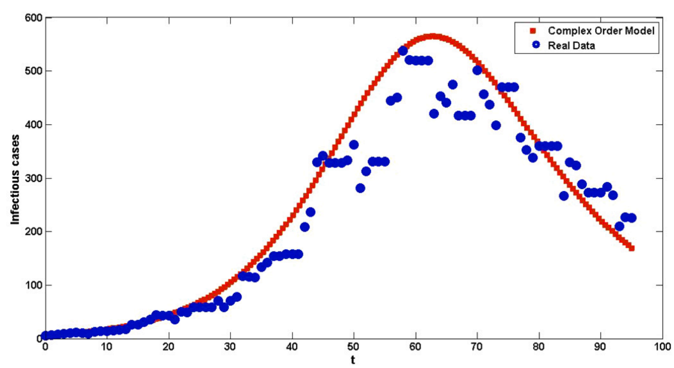

Fig. 1. Numerical simulation of Infectious Cases at $0.98 \pm 0.03i$ using NS2LIM.

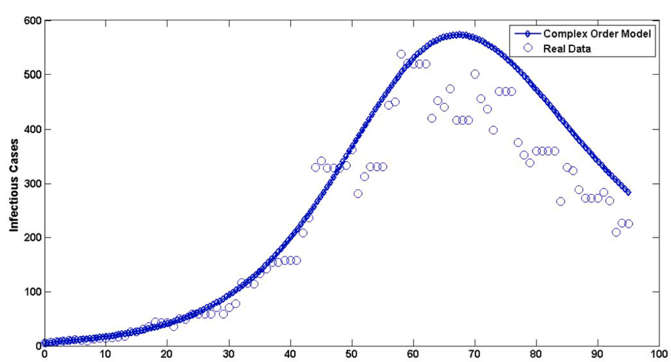

Fig. 2. Numerical simulation of Infectious Cases at $0.98 \pm 0.1i$ using NS2LIM.

Fig. 1, Fig. 2, Fig. 3 and Fig. 4. The fitting of the 7-days moving average data of monkeypox in United States from June 13 to Sep 16, 2022 is given in Fig. 1 at $0.98 \pm 0.03i$, in Fig. 2 at $0.98 \pm 0.1i$, in Fig. 3 at $0.98 \pm 0.2i$ and Fig. 4 at $0.98 \pm 0i$. The Fig. 5shows how the behaviour of the solutions of exposed humans class, infected humans class, isolation humans class, exposed rodent class and infected rodent class changes according to integer, fractional and complex order derivatives, its clear that a new behaviour appears using complex order derivatives. Then complex order model gave us more details about spreading the disease. Moreover, this model is more general than the integer and fractional model. Fig. 6 shows the numerical simulations of infected humans class and infected rodent class at $0.9 + 0i$ and $0.9 + 0.2i$, we observed that new behaviour appears in complex order derivatives. Fig. 7 shows the numerical simulations of infected rodent at $0.9 + 0i$ and $0 + 0.3i$, this result at $0 + 0.3i$ don't appear at fractional and integer order derivatives. Fig. 8 shows the numerical simulations of exposed humans class and infected humans class at $0.9 + 0.4i$ and $0.9 + 0.5i$. The

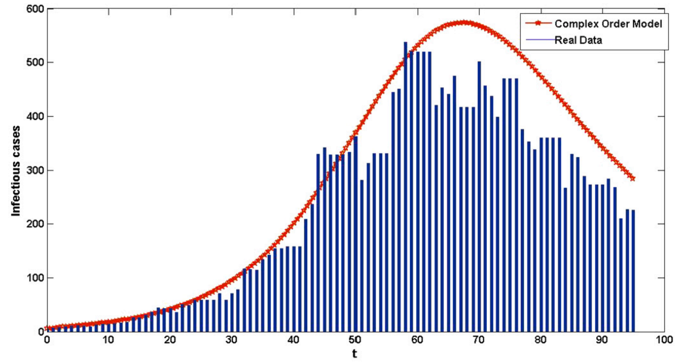

Fig. 3. Numerical simulation of Infectious Cases at $0.98 \pm 0.2i$ using NS2LIM.

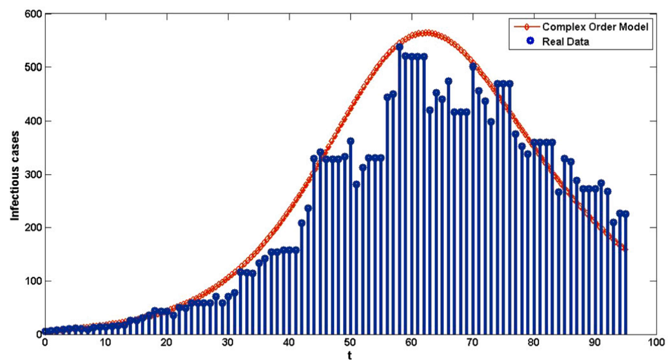

Fig. 4. Numerical simulation of Infectious Cases at $0.98 \pm 0i$ using NS2LIM.

Fig. 9 shows the numerical simulations of exposed rodent class and humans rodent class at $0.8 + 0.5i$ and $0.8 + 0.6i$ respectively. Fig. 10 shows the numerical simulations of exposed humans class and infected humans class at $0.9 + 0i$ and $0.9 + 0.5i$ respectively.

8. Conclusions

The mathematical model for monkeypox complex order is discussed in this article. It is easier to depict biological processes with memory using this dynamical model. The complex-order system also exhibits a diverse range of dynamics for the complex-order derivative value. To get numerical solutions for a complex order monkeypox model with Mittag-Leffler kernels, the numerical methods based on Lagrange polynomial interpolation are used. When the imaginary part of the complex order is equal to zero, the complex order derivative is recognised as a popularisation of the fractional order derivative and the integer order derivative. In the case of the real part in complex order derivatives is zero, the new behaviour appears that doesn't appear in integer and fractional order derivative. The complex order derivatives have numerous advantages, including rich dynamics and the ability to provide fresh light on the modelling of intracellular delay by changing the complex order derivative value. We study the duration and monkeypox outbreak's transmission pattern in the United States and present numerical simulations to illustrate our findings. The approach is precise, effective, and straightforward. The future directions of this work are developing new methods to study the proposed model and compare the obtained results with previous studies. Optimal control problem using direct methods and derive the necessary and sufficient optimality conditions is the second target.

Declaration of competing interest

The authors declare that they have no conflict of interests.

References

[1] U.S. Monkeypox, Case trends reported to CDC, https://www.cdc.gov/poxvirus/monkeypox/response/2022/mpx-trends.html.
[2] J. Breman, Monkeypox: an emerging infection for humans, Emerg. Infect. 4 (2000) 45–67.
[3] P. Kumar, M. Vellappandi, Z. Khan, S. Sivalingam, A. Kaziboni, V. Govindaraj, A case study of monkeypox disease in the United States using mathematical modeling with real data, Math. Comput. Simul. 444 (465) (2023) 213.
[4] R. Levine, A. Peterson, K. Yorita, D. Carroll, I. Damon, M. Reynolds, Ecological niche and geographic distribution of human monkeypox in Africa, PLoS ONE 2 (1) (2007) e176.
[5] L. Leandry, E. Mureithi, An investigation on the monkeypox virus dynamics in human and rodent populations for a deterministic mathematical model, Inform. Med. Unlocked 41 (2023) 101325.
[6] S. Okyere, J. Ackora-Prah, Modeling and analysis of monkeypox disease using fractional derivatives, Results Eng. 100786 (2023) 17.
[7] D. Bisanzio, R. Reithinger, Projected burden and duration of the 2022 monkeypox outbreaks in non-endemic countries, The Lancet. Microbe (2022).
[8] J. Riopelle, V. Munster, J. Port, Atypical and unique transmission of monkeypox virus during the 2022 outbreak, an overview of the current state of knowledge, Viruses 14 (9) (2022).

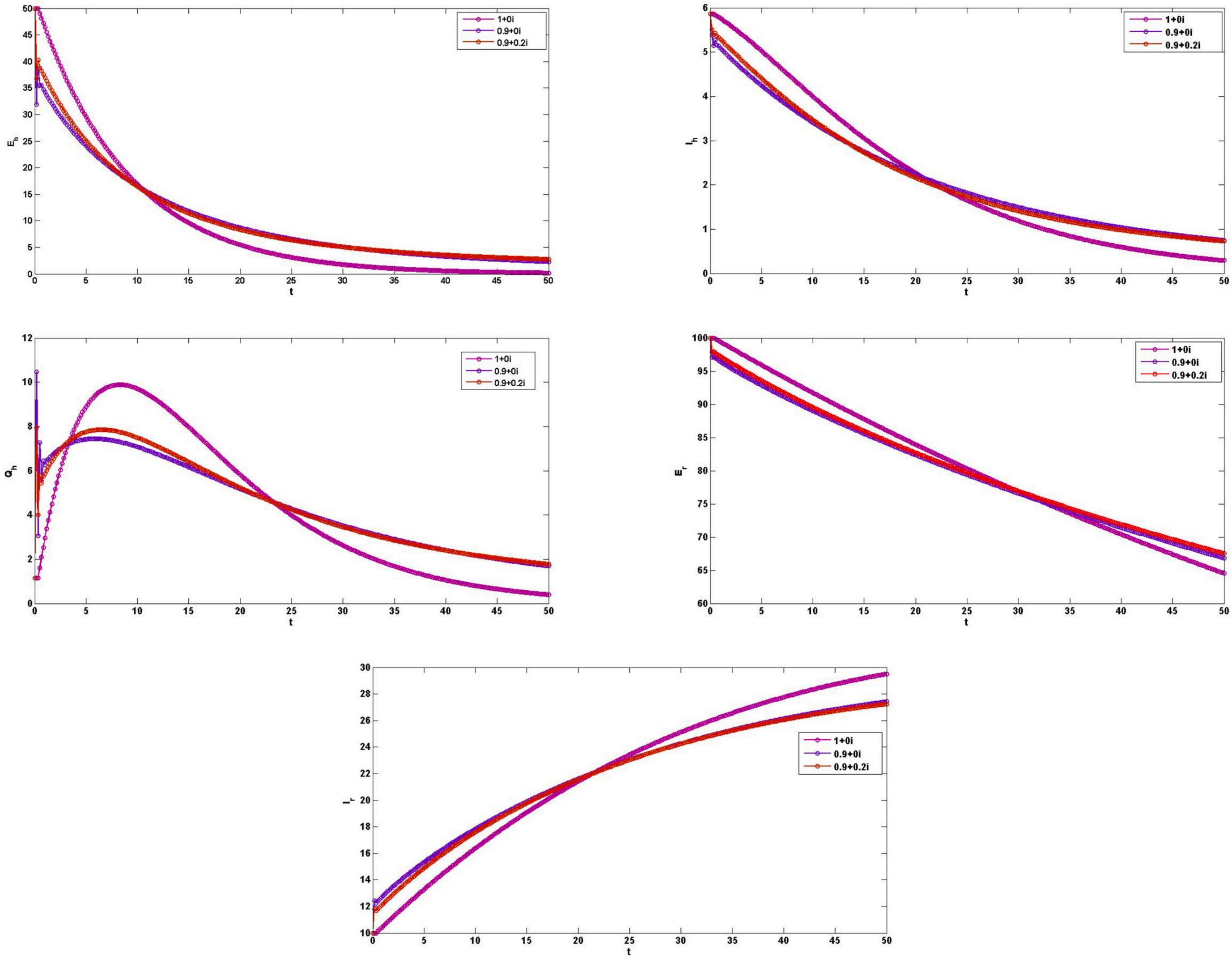

Fig. 5. Numerical simulation of variables E_h, I_h, Q_h, E_r, I_r of monkeypox model using NS2LIM.

[9] N. Kumar, A. Acharya, H. Gendelman, S. Byrareddy, The 2022 out-break and the pathobiology of the monkeypox virus, Autoimmunity 102855 (2022).
[10] S. Usman, I. Adamu, Modeling the transmission dynamics of the monkeypox virus infection with treatment and vaccination interventions, Appl. Math. Phys. 5 (12) (2017) 2335.
[11] A. Khan, Y. Sabbar, A. Din, Stochastic modeling of the monkeypox 2022 epidemic with cross-infection hypothesis in a highly disturbed environment, Math. Biosci. Eng. 19 (12) (2022) 13560–13581.
[12] R. Grant, L. Nguyen, R. Breban, Modelling human-to-human transmission of monkeypox, Bull. World Health Organ. 98 (9) (2020) 638.
[13] S. Bankuru, S. Kossol, W. Hou, P. Mahmoudi, J. Rycht, D. Taylor, A game-theoretic model of monkeypox to assess vaccination strategies, PeerJ 8 (2020) e9272.
[14] O. Peter, F. Oguntolu, M. Ojo, A. Oyeniyi, R. Jan, I. Khan, Fractional order mathematical model of monkeypox transmission dynamics, Phys. Scr. 97 (8) (2022) 084005.
[15] O. Peter, S. Kumar, N. Kumari, F. Oguntolu, K. Oshinubi, R. Musa, Transmission dynamics of monkeypox virus: a mathematical modelling approach, Model. Earth Syst. Environ. 8 (2022) 3423–3434.
[16] J. Solis-Perez, J. Gomez-Aguilar, Novel numerical method for solving variable-order fractional differential equations with power, exponential and Mittag-Leffler laws, Chaos Solitons Fractals 114 (2018) 175–185.
[17] N. Sweilam, Z. Mohammed, W. Abd kareem, Numerical treatments for a multi-time delay complex order mathematical model of HIV/AIDS and malaria, Alex. Eng. J. 61 (12) (2022) 10263–10276.
[18] N. Sweilam, S. AL-Mekhlafi, Z. Mohammed, D. Baleanu, Optimal control for variable order fractional HIV/AIDS and malaria mathematical models with multi-time delay, Alex. Eng. J. 59 (5) (2020) 3149–3162.
[19] C. Pinto, A. Carvalho, Fractional complex-order model for HIV infection with drug resistance during therapy, J. Vib. Control 22 (9) (2016) 2222–2239.
[20] A. Fernandez, A complex analysis approach to Atangana-Baleanu fractional calculus, J. Math. Methods Appl. Sci. 1 (18) (2019).
[21] N. Sweilam, F. Rihan, S. AL-Mekhlafi, A fractional-order delay differential model with optimal control for cancer treatment based on synergy between anti-angiogenic and immune cell therapies, Discrete Contin. Dyn. Syst. 13 (9) (2020) 2403–2424.
[22] A. Neamaty, M. Yadollahzadeh, R. Darzi, On fractional differential equation with complex order, Prog. Fract. Diff. Appl. 223–227 (2015).
[23] N. Sweilam, S. AL-Mekhlafi, A. Alshomrani, D. Baleanu, Comparative study for optimal control nonlinear variable-order fractional tumor model, Chaos Solitons Fractals 136 (2020).
[24] E. Love, Fractional derivatives of imaginary order, J. Lond. Math. Soc. 3 (2) (1971) 241–259.
[25] N. Sweilam, F. Rihan, S. AL-Mekhlafi, A fractional-order delay differential model with optimal control for cancer treatment based on synergy between anti-angiogenic and immune cell therapies, Discrete Contin. Dyn. Syst. Ser. S 13 (9) (2020) 2403–2424.
[26] N. Sweilam, S. Al-Mekhlafi, T. Assiri, A. Atangana, Optimal control for cancer treatment mathematical model using Atangana-Baleanu-Caputo fractional derivative, Adv. Cont. Discr. Mod. 1 (2020) 1–21.
[27] A. Atangana, D. Baleanu, New fractional derivatives with non-local and non-singular kernel theory and application to heat transfer model, Therm. Sci. 20 (2) (2016) 763–769.
[28] M. Khader, N. Sweilam, A. Mahdy, N. Abdel Moniem, Numerical simulation for the fractional SIRC model and influenza, Appl. Math. Inf. Sci. 8 (3) (2014) 1029–1036.

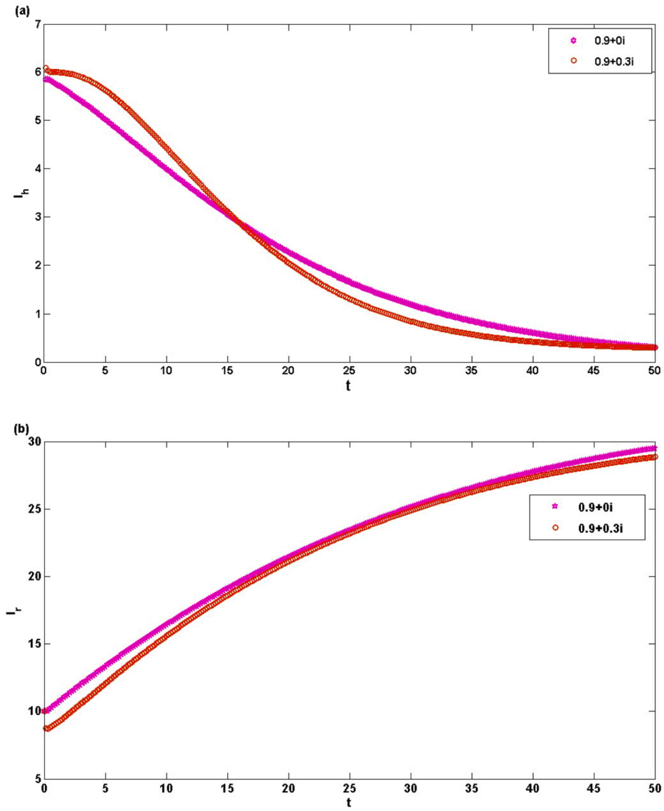

Fig. 6. Numerical simulation of (a) Infected humans class and (b) Infected rodent class at different complex values using NS2LIM.

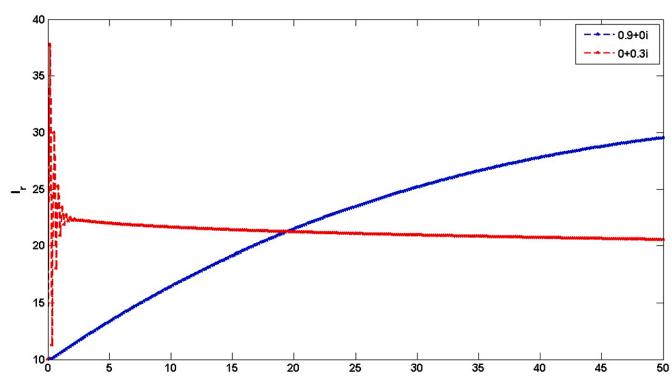

Fig. 7. Numerical simulation of Infected rodent at different complex values using NS2LIM.

[29] N. Sweilam, M. Khader, M. Adel, Numerical simulation of fractional Cable equation of spiny neuronal dendrites, J. Adv. Res. 5 (2) (2014) 253–259.
[30] A. Atangana, K. Owolabi, New numerical approach for fractional differential equations, Math. Model. Nat. Phenom. 13 (1) (2018) 1–17.
[31] D. Valerio, J. Costa, Variable-order fractional derivatives and their numerical approximations, in: Symposium on Fractional Signals and Systems, 2009, pp. 4–9.
[32] A. Carvalho, M. Pinto, A delay fractional order model for the co-infection of malaria and HIV/AIDS, Int. J. Dyn. Control 5 (2017) 168–186.
[33] R. Mickens, Nonstandard Finite Difference Models of Differential Equations, World Scientific, Singapore, 2005.
[34] R. Mickens, Calculation of denominator functions for nonstandard finite difference schemes for differential equations satisfying a positivity condition, Numer. Methods Partial Differ. Equ. 23 (2007) 672–691.
[35] K. Patidar, Nonstandard finite difference methods: recent trends and further developments, J. Differ. Equ. Appl. 22 (6) (2016) 817–849.
[36] N. Sweilam, I. Soliman, S. AL-Mekhlafi, Nonstandard finite difference method for solving the multi-strain TB model, J. Egypt. Math. Soc. 25 (2) (2017) 129–138.
[37] N. Sweilam, S. AL-Mekhlafi, Optimal control for a nonlinear mathematical model of tumor under immune suppression: a numerical approach, Optim. Control Appl. Methods 1 (16) (2018).
[38] N. Sweilam, S. AL-Mekhlafi, D. Baleanu, Optimal control for a fractional tuberculosis infection model including the impact of diabetes and resistant strains, Adv. Res. 17 (2019) 125–137.
[39] M. Toufik, A. Atangana, New numerical approximation of fractional derivative with non-local and non-singular kernel, application to chaotic models, Eur. Phys. J. Plus 132 (10) (2017) 1–16.
[40] N. Sweilam, S. AL-Mekhlafi, Nonstandard finite difference method for solving complex-order fractional Burgers' equations, Adv. Res. 25 (2020) 19–29.
[41] R. Abazari, M. Ganji, Extended two-dimensional DTM and its application on nonlinear PDEs with proportional delay, Int. J. Comput. Math. 88 (8) (2011) 1749–1762.
[42] N. Sweilam, W. Abd kareem, S. AL-Mekhlafi, Z. Mohammed, Numerical treatments for a complex order fractional HIV infection model with drug resistance during therapy, Prog. Fract. Differ. Appl. 1 (1) (2020) 1–13.

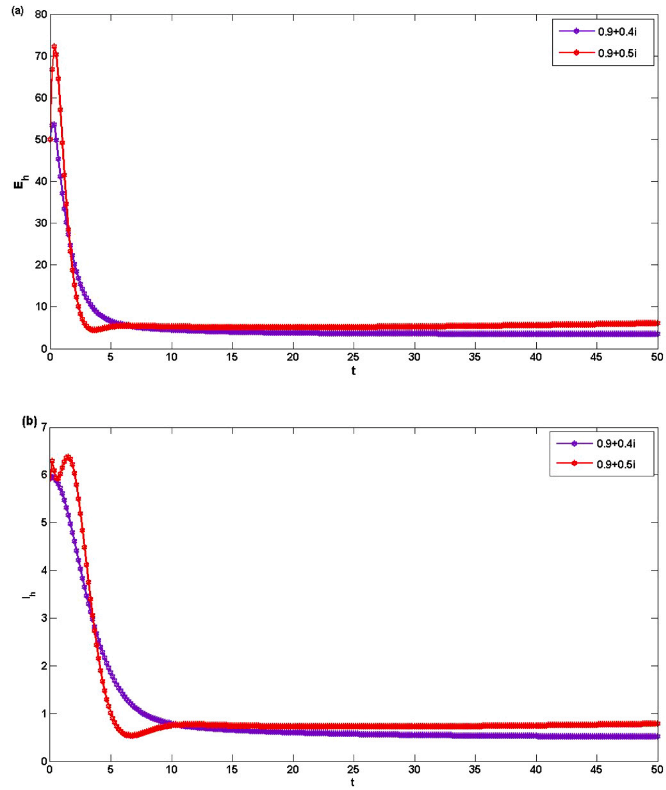

Fig. 8. Numerical simulation of (a) Exposed humans and (b) Infected humans at different complex values using NS2LIM.

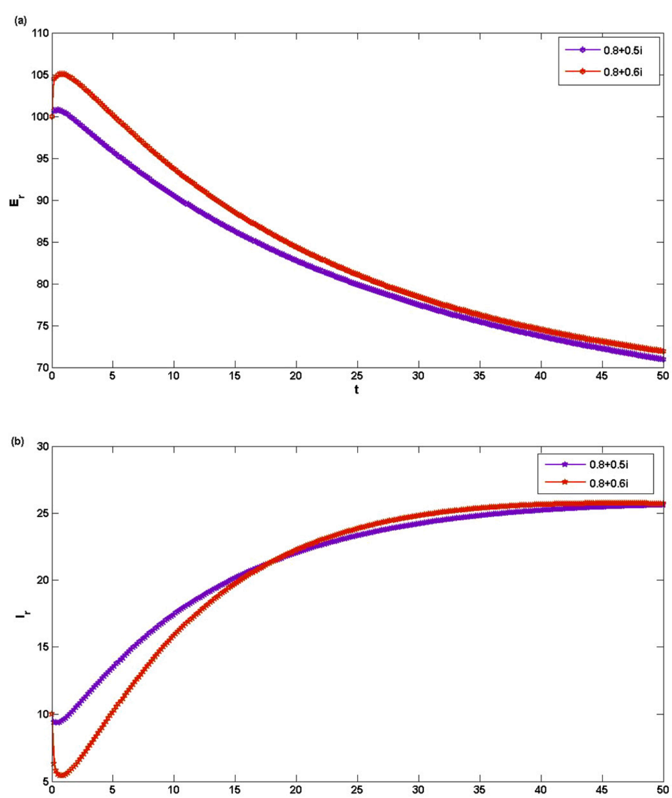

Fig. 9. Numerical simulation of (a) Exposed rodent and (b) Infected rodent at different complex values using NS2LIM.

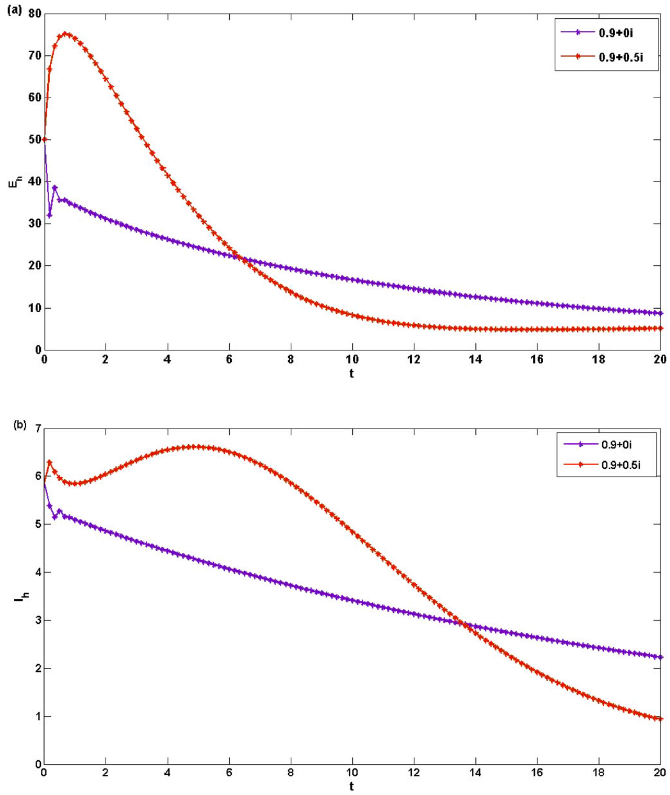

Fig. 10. Numerical simulation of (a) Exposed humans and (b) Infected humans at different complex values using NS2LIM.

[43] M. Akrami, K. Owolabi, On the solution of fractional differential equations using Atangana's beta derivative and its applications in chaotic systems, Sci. Afr. 21 (2023) e01879.
[44] A. Atangana, A. Akgul, K. Owolabi, Analysis of fractal fractional differential equations, Alex. Eng. 59 (2) (2020) 1117–1134.
[45] B. Karaagac, K. Owolabi, E. Pindza, Analysis and new simulations of fractional noyes-field model using Mittag-Leffler kernel, Sci. Afr. 17 (3) (2022) e01384.
[46] P. Naik, K. Owolabi, J. Zu, M. Naik, Modeling the transmission dynamics of Covid-19 pandemic in Caputo type fractional derivative, Multiscale Modell. 12 (2021).
[47] K. Owolabi, A. Shikongo, Fractal fractional operator method on HER2+ breast cancer dynamics, Appl. Comput. Math. 7 (3) (2021).
[48] K. Owolabi, Analysis and simulation of herd behaviour dynamics based on derivative with nonlocal and nonsingular kernel, Results Phys. 22 (2021) 103941.
[49] B. Karaagac, K. Owolabi, K. Nisar, Analysis and dynamics of illicit drug use described by fractional derivative with Mittag-Leffler kernel, Tech Science Press 65 (3) (2020) 1905–1924.